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Container cap for storing specimen

Abstract

The present invention can provide a container cap for storing a specimen, the container cap being characterized by comprising: a cap body which includes a tubular side portion and a bottom portion coupled to one end of the side portion, and is detachably screw-fastened to an open side of a specimen storage container in which one or more specimen collection swabs are inserted; an insertion coupling part which protrudes from the bottom portion toward the specimen storage container and has an inner space formed therein, wherein respective stick portions of the one or more specimen collection swabs are inserted into the inner space and fitted and coupled thereto; and a protrusion fitting part which is positioned at the central portion of the inner space and protrudes from the bottom surface of the cover body.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2835246	12/1957	Boettger	600/572	A61F 13/38
4409988	12/1982	Greenspan	435/304.2	B01L 3/5029
4803998	12/1988	Kezes	600/572	A61F 13/38
5458113	12/1994	Burns	422/547	A61B 5/150343
5543115	12/1995	Karakawa	436/66	A61B 10/0038
5830154	12/1997	Goldstein	600/572	B01L 3/5029
5965453	12/1998	Skiffington	435/287.7	C12Q 1/66
6780160	12/2003	Zhou	600/562	A61B 10/0096
7323144	12/2007	Arai	422/534	B01L 3/5021
7517497	12/2008	Price	422/547	B26F 1/16
7993871	12/2010	Skiffington	435/287.4	G01N 1/02
8075850	12/2010	Sangha	422/406	B01L 99/00
8623665	12/2013	Poll	436/177	A61B 10/0096
8728414	12/2013	Beach	422/549	A61B 10/02
8961896	12/2014	McSherry	600/572	C12M 1/30
9027420	12/2014	Ward	73/864.71	G01N 1/02
10004444	12/2017	Bauer	N/A	A61B 5/150343
11325129	12/2021	Staton	N/A	B65D 1/36
11709175	12/2022	Hopper	435/6.12	B01L 3/502
D1042873	12/2023	Goh	D24/119	N/A
D1042879	12/2023	Goh	D24/224	N/A
2004/0019295	12/2003	Zhou	600/562	A61B 10/0096
2004/0019298	12/2003	Zhou	600/564	A61B 10/0096
2005/0252820	12/2004	Sanchez-Felix	206/569	A61B 10/0045

2006/0057027	12/2005	Hudak	422/549	A61B 10/0051
2006/0115385	12/2005	Jon Meyer	422/547	A61B 10/0096
2008/0206740	12/2007	Skiffington	435/5	B01L 3/5029
2009/0156962	12/2008	Yong	600/569	A61B 10/0045
2010/0043574	12/2009	Katsumata	422/400	B01L 3/50825
2012/0009588	12/2011	Rajagopal	435/7.1	C12Q 1/04
2012/0094303	12/2011	Engel	435/7.1	G01N 33/54388
2012/0141341	12/2011	Bartfeld	422/549	B01L 3/50825
2012/0220042	12/2011	Sangha	422/411	G01N 1/02
2012/0220043	12/2011	Sangha	422/411	G01N 1/02
2013/0216453	12/2012	Eldridge	422/550	A61J 1/1412
2013/0302219	12/2012	Li	422/550	B01L 3/50825
2014/0017147	12/2013	Kim	422/501	B01L 3/50825
2017/0036205	12/2016	Bishop	N/A	A61B 10/0045
2017/0071582	12/2016	Buck	N/A	A61B 10/0051
2017/0260563	12/2016	Lappalainen	N/A	G01N 33/528
2020/0056965	12/2019	Gao	N/A	G01N 1/02
2020/0155125	12/2019	Ouyang	N/A	C12M 1/30
2022/0338848	12/2021	Martello	N/A	B01L 3/5029
2023/0172591	12/2022	Robbins	600/573	B01L 3/5029
2023/0311130	12/2022	Reber	422/600	B01L 7/52
2023/0355219	12/2022	Ling	N/A	A61B 10/0045
2023/0363744	12/2022	Goh	N/A	A61B 10/0051
2023/0397856	12/2022	Walker	N/A	A61B 5/150022
2024/0042449	12/2023	Hassberg	N/A	B01L 3/523
2024/0044750	12/2023	Goh	N/A	A61B 10/02
2024/0188942	12/2023	Johnson	N/A	A61B 10/0051
2024/0261788	12/2023	Goh	N/A	A61B 10/0096
2024/0270457	12/2023	Goh	N/A	B01L 3/50825
2024/0280449	12/2023	Goh	N/A	B01L 3/50825
2025/0033036	12/2024	Lei	N/A	B01L 3/5023

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
5584287	12/2013	JP	N/A
20110006453	12/2010	KR	N/A
101140751	12/2011	KR	N/A
101284703	12/2012	KR	N/A
101309066	12/2012	KR	N/A
20200000438	12/2019	KR	N/A

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a container cover for specimen storage, and more particularly to a container cover for specimen storage capable of more efficiently storing and transporting a specimen collected from a subject in order to identify or test for infection with various kinds of communicable diseases.

BACKGROUND ART

(2) Due to the recent spread of COVID 19 (Corona Virus Disease 19), testing for infection has been actively conducted, and testing is done by collecting a specimen from a subject.

(3) In contrast, for common respiratory diseases, a swab for specimen collection is inserted into the nasal cavity or oral cavity of a subject, secretions are collected from the mucous membrane of the nasal cavity or oral cavity lining, and the secretions are examined to test for respiratory diseases.

(4) Although recently prevalent COVID 19 is a respiratory disease, secretions are collected from both the mucous membranes of the nasal cavity and the oral cavity lining of the subject in order to improve accuracy of the test. After specimens are collected, collection sticks are placed in a specimen preservation container containing a preservation solution (medium), and the specimen preservation container is stored or transported to a testing site.

(5) Meanwhile, a conventional specimen collection and storage kit configured to collect and store a specimen includes a swab configured to collect a specimen, a specimen container configured to receive a medium and to allow the swab having the collected specimen to be inserted therein so as to be stored therein, and a container cover for specimen storage configured to seal the specimen container.

(6) However, the conventional specimen collection and storage kit is configured to allow only one swab to be inserted therein and stored therein. When specimens are collected from two or more body parts depending on the type of test, therefore, swabs having the collected specimens must be separately stored in different specimen collection and storage kits even though the specimens are from the same subject.

(7) Because of this, it is difficult to store and manage the conventional specimen collection and storage kits due to requirement of more specimen collection and storage kits as well as the laborious task of repeatedly filling in information of the same subject on the respective specimen containers.

(8) In addition, the conventional specimen collection and storage kits require an inspector to manually remove the swab having the collected specimen from the specimen container after transportation thereof to the testing site, which is not only cumbersome, but may also lead to unnecessary contamination during the swab removal process.

DISCLOSURE

Technical Problem

(9) It is an object of the present invention to provide a container cover for specimen storage that allows a plurality of specimen collection swabs to be stored in a single specimen collection container when specimens are collected from two or more body parts of a subject, thereby improving workability and facilitating storage and management.

(10) It is another object of the present invention to provide a container cover for specimen storage configured such that it is unnecessary for an inspector to perform a separate operation to remove a specimen collection swab from a specimen container, whereby it is possible to eliminate conventional inconvenience of removing the specimen collection swab and to prevent unnecessary contamination that may occur during the operation to remove the swab.

Technical Solution

(11) The present invention provides a container cover for specimen storage, the container cover including a cover body comprising a tubular side portion and a bottom portion coupled to one end of the side portion, the cover body being detachably screwed to open one side of a specimen

preservation container in which one or more specimen collection swabs are provided by insertion, an insertion coupling portion formed so as to protrude from the bottom portion toward the specimen preservation container, the insertion coupling portion having an inner space defined therein such that stick portions of the specimen collection swabs can be inserted and fitted into and coupled to the inner space, and a protruding fitting portion located in the center of the inner space, the protruding fitting portion being formed so as to protrude from the bottom portion of the cover body, the protruding fitting portion being configured such that the stick portions are fitted and coupled between an outer surface of the protruding fitting portion and an inner surface of the insertion coupling portion, wherein, when the specimen preservation container is coupled to the container cover, the stick portions are fitted into and coupled to an arbitrary position in the inner space.

Advantageous Effects

(12) In a container cover for specimen storage according to the present invention, it is possible to store a plurality of swabs having specimens collected from two or more body parts of a subject in a single specimen collection kit, whereby it is possible to dramatically reduce tasks, such as filling in information of the subject, and therefore it is possible to improve workability, and it is possible to reduce the number of specimen collection and storage kits, and therefore it is possible to reduce a space necessary for storage and to achieve easier storage and management.

(13) In addition, the container cover for specimen storage according to the present invention is configured such that, when the container cover for specimen storage is removed, specimen collection swabs are also removed from a specimen preservation container without the need for an inspector to remove the specimen collection swabs from the specimen preservation container using tweezers, whereby it is possible to prevent unnecessary contamination, and the need to remove the specimen collection swabs and to disinfect tools such as the tweezers is eliminated, whereby it is possible to eliminate operational inconvenience and to improve workability.

(14) Also, in the container cover for specimen storage according to the present invention, it is possible to secure the force of fastening to the specimen preservation container and to improve airtightness and sealing force through an airtight portion, whereby it is possible to effectively prevent leakage of a specimen medium and contamination from the outside.

(15) The container cover for specimen storage according to the present invention is simple in structure and has a reduced number of parts, whereby it is possible to reduce manufacturing costs, which is economical.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an exterior view showing the state in which a container cover for specimen storage according to an embodiment of the present invention is coupled to a container.

(2) FIG. 2 is an exploded perspective view showing the container cover for specimen storage according to the embodiment of the present invention and the container.

(3) FIG. 3 is a sectional view showing the internal structure in the state in which the container cover for specimen storage according to the embodiment of the present invention is coupled to the container.

(4) FIG. 4 is a sectional view showing a detailed configuration of the container cover for specimen storage according to the embodiment of the present invention.

(5) FIG. 5 is a sectional view taken along line V-V of FIG. 3.

(6) FIG. 6 is a plan sectional view showing another embodiment of the container cover for specimen storage according to the embodiment of the present invention in which a contact recess is formed in a protruding fitting portion.

- (7) FIG. 7 is a sectional view showing a coupling structure between the container cover for specimen storage according to the embodiment of the present invention and a specimen collection swab.
- (8) FIG. 8 is an enlarged view of part “A” of FIG. 7.
- (9) FIGS. 9 to 11 are sectional views showing insertion of the specimen collection swab and the operation of a separation prevention portion in response to rotation of the container cover for specimen storage according to the embodiment of the present invention of FIG. 6.
- (10) FIG. 12 is a sectional view showing the container cover for specimen storage according to the embodiment of the present invention and a specimen preservation container.
- (11) FIG. 13 is an enlarged view of part “B” of FIG. 12.
- (12) FIG. 14 is a sectional view showing the state in which the container cover for specimen storage according to the embodiment of the present invention is coupled to the specimen preservation container in tight contact therewith as the result of rotation of the container cover in FIG. 12.
- (13) FIG. 15 is an enlarged view of part “C” of FIG. 14.
- (14) FIG. 16 is a sectional view showing a further embodiment of the container cover for specimen storage according to the embodiment of the present invention.
- (15) FIG. 17 is a view showing a process of using the container cover for specimen storage according to the embodiment of the present invention.

BEST MODE

- (16) The present invention may be changed in various manners and may have various embodiments, wherein specific embodiments will be described with reference to the drawings. However, the present invention is not limited to the specific embodiments, and it should be understood that the present invention includes all modifications, equivalents, or substitutions included in the idea and technical scope of the present invention.
- (17) Although terms including ordinal numbers, such as “first” and “second,” may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another component. For example, within the scope defined by the present invention, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element. The term “and/or” includes any and all combinations of one or more of the associated listed items.
- (18) It should be understood that, when a component is referred to as being “connected to” or “coupled to” another component, it may be directly connected to or coupled to another component or intervening components may be present. In contrast, it should be understood that, when a component is referred to as being “directly connected to” or “directly coupled to” another component, there are no intervening components present.
- (19) The terms used in the present application are provided only to described specific embodiments, and do not limit the present invention. Singular forms are intended to include plural forms as well, unless the context clearly indicates otherwise. In the present application, it should be understood that the terms “includes,” “has,” etc. specify the presence of stated features, numbers, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, elements, components, or combinations thereof.
- (20) Unless otherwise defined, all terms, including technical and scientific terms, used in this specification have the same meanings as those commonly understood by a person having ordinary skill in the art to which the present invention pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having meanings consistent with their meanings in the context of the relevant art and the present disclosure, and are not to be interpreted in an idealized or overly formal sense unless expressly so defined herein. Also, in describing the present invention, identical reference numerals are used for identical components

in the drawings, and duplicate descriptions of identical components will be omitted for ease of overall understanding.

(21) A specimen collection and storage kit according to an embodiment of the present invention is configured such that, when specimens are collected from two or more body parts of a subject, specimen collection swabs **100** can be stored in a single specimen preservation container **200**, such that it is possible to prevent contamination from the outside during storage and transportation due to good airtightness, and such that the specimen collection swabs **100** are removed from the specimen preservation container **200** together with a container cover **300** for specimen storage at the time of inspection without the need to remove the specimen collection swabs **100** from the specimen preservation container **200**.

(22) Hereinafter, preferred embodiments of the present invention will be described in more detail with reference to the accompanying drawings.

(23) A container cover **300** for specimen storage according to an embodiment of the present invention is configured to be included in a specimen collection and storage kit, and the specimen collection and storage kit will first be described before describing the container cover **300** for specimen storage according to the present invention.

(24) FIGS. **1** and **2** are views showing a specimen collection and storage kit, wherein the specimen collection and storage kit includes a specimen collection swab **100**, a specimen preservation container **200**, and a container cover **300** for specimen storage according to the present invention.

(25) First, the specimen collection swab **100**, which is configured to collect a specimen from a subject, may be formed in the shape of a swab, as shown.

(26) The specimen collection swab **100** may include a head portion **110** configured to collect a specimen and a stick portion **120** coupled to the head portion **110**.

(27) The head portion **110** may have a roughly spherically rounded shape, and may be made of a material, such as a fiber. That is, the head portion may be configured to have a shape and material that facilitate collection of a specimen from a subject.

(28) The stick portion **120** may be formed in the shape of a stick, and may be configured such that one end of the stick portion is coupled to the head portion **110** and the other end of the stick portion can be gripped by an inspector.

(29) The stick portion **120** may be formed to have a predetermined long length to facilitate insertion into the nasal cavity or oral cavity in order to easily collect a specimen.

(30) The stick portion **120** may be provided with an incision (not shown) configured such that a part of incision can be cut by the inspector. When a specimen is collected, therefore, the stick portion **120** may be maintained long to facilitate specimen collection, and when the stick portion is inserted into the specimen preservation container **200** after specimen collection, the stick portion is cut so as to correspond to the length of the specimen preservation container **200**, whereby it is possible to easily achieve shielding of the container cover **300** for specimen storage.

(31) The specimen preservation container **200** may be formed in the shape of a cup shielded at one side, a receiving space **210** may be formed in the specimen preservation container such that a medium **1** and the specimen collection swab **100** are received and stored in the receiving space **210**, and the specimen preservation container may be provided at the other open side thereof with an insertion hole **230**, through which the specimen collection swab **100** is inserted.

(32) Referring to FIG. **3**, the specimen preservation container **200** may be configured such that the shielded one-side bottom surface of the specimen preservation container is formed in a concave shape, as shown.

(33) The specimen preservation container **200** may be made of a transparent material to allow the inspector to view the interior or an opaque material.

(34) The specimen preservation container **200** may be made of, but is not limited to, a synthetic resin-based material that is easy to manufacture and economical.

(35) Hereinafter, a container cover **300** for specimen storage according to an embodiment of the

present invention will be described.

(36) The container cover **300** for specimen storage according to the embodiment of the present invention may be detachably screwed to the open insertion hole **230** side of the specimen preservation container **200** to open and close the receiving space **210** such that the specimen collection swab **100** can be stored and transported.

(37) Furthermore, the container cover **300** for specimen storage according to the embodiment of the present invention may be coupled to the one or more specimen collection swabs **100** when coupled to the specimen preservation container **200** such that, when separated from the specimen preservation container **200**, the specimen collection swabs **100** are also removed from the specimen preservation container **200**.

(38) Referring to FIG. **4**, the container cover **300** for specimen storage includes a cover body **350**, an insertion coupling portion **400**, a protruding fitting portion **500**, a separation prevention portion **700**, and an airtight portion **800**.

(39) The cover body **350** includes a tubular side portion **310** coupled to an outer circumferential surface of one side of the specimen preservation container **200** and a bottom portion **320** coupled to one end of the side portion **310** to shield the insertion hole **230**.

(40) Here, the insertion coupling portion **400** and the protruding fitting portion **500** may be coupled to the bottom portion **320** toward a lower surface facing the specimen preservation container **200**, i.e., the receiving space **210**.

(41) In the container cover **300** for specimen storage, an interstitial space **330** is defined between an inner circumferential surface of the side portion **310** and an outer circumferential surface of the insertion coupling portion **400**, and one end of the specimen preservation container **200** may be inserted into and coupled to the interstitial space **330**.

(42) The interstitial space **330** may be formed such that the width of the interstitial space is gradually reduced from an inlet, through which the specimen preservation container **200** is inserted, to the bottom portion **320**, whereby easy insertion of the specimen preservation container **200** may be achieved, and the insertion end of the specimen preservation container **200** and the container cover **300** for specimen storage may come into tighter contact with each other.

(43) The container cover **300** for specimen storage may be detachably coupled to the specimen preservation container **200** via a detachment means **600**. A screw coupling structure capable of securing fastening force and airtightness may be applied the detachment means **600**, as shown.

(44) The detachment means **600** includes a first screw thread **610** formed on an outer circumferential surface of an upper end of the specimen preservation container **200** and a second screw thread **620** formed on an inner surface of the side portion **310** of the container cover **300** for specimen storage so as to be engaged with the first screw thread **610**.

(45) Here, although a screw coupling structure is applied to the detachment means **600** in a preferred embodiment, various other detachment structures, such as a snap structure, may be used as long as it is possible to secure fastening force and airtightness.

(46) The insertion coupling portion **400** is formed so as to protrude from the bottom portion **320** of the cover body **350** toward the receiving space **210**, an inner space **410** is formed in the insertion coupling portion, and the stick portions **120** of the one or more specimen collection swabs **100** may be inserted and fitted into the inner space **410**.

(47) The insertion coupling portion **400** may be formed so as to have an inclined surface **420** formed at an exposed end and inclined downward from the outer surface to the inner surface thereof. The inclined surface **420** serves to guide the stick portion **120** such that stick portion can be more easily inserted into the inner space **410** without being caught by the end.

(48) Referring to FIG. **5**, the insertion coupling portion **400** may have the shape of a cylindrical tube and may be formed in the shape of a closed loop in plan section, whereby the ends of the stick portions **120** may be fitted into and received at an arbitrary position in a cylindrical inner space **410**.

(49) Specifically, when the container cover **300** for specimen storage is coupled to the specimen preservation container **200** in the state in which a plurality of specimen collection swabs **100** is disposed in the specimen preservation container **200** by insertion, the stick portion **120** may be fitted into and coupled to an arbitrary position in the inner space **410** of the insertion coupling portion **400** without the need to separately set the coupling position of the stick portion **120**.

(50) The planar insertion area of the internal space **410** of the insertion coupling portion **400** may be variously set by considering the diameter and maximum number of the stick portions **120** to be inserted into the internal space **410**.

(51) The protruding fitting portion **500** may be located in the center of the inner space **410**, and may be formed so as to protrude from a lower surface of the container cover **300** for specimen storage, i.e., the bottom portion **320**, toward the receiving space **210**.

(52) The one or more stick portions **120** may be fitted into and coupled to an arbitrary position between the outer surface of the protruding fitting portion **500** and the inner surface of the insertion coupling portion **400**, and the number of the stick portions **120** that can be fitted thereto and coupled thereto may optionally be set to one to a plurality.

(53) Diameters and positions may be set such that the distance between the outer circumferential surface of the protruding fitting portion **500** and the inner circumferential surface of the insertion coupling portion **400** corresponds to the diameter of the stick portion **120**, whereby the stick portion **120** is fitted between the outer circumferential surface of the protruding fitting portion **500** and the inner circumferential surface of the insertion coupling portion **400**.

(54) On the assumption that the stick portions **120** have the same diameter, the protruding fitting portion **500** may be located on the center axis of the inner space **410**, i.e., in the center of the inner space **410** in plan, as shown.

(55) The protruding fitting portion **500** may be formed in a cylindrical shape, as shown; however, the present invention is not limited thereto.

(56) As described above, the protruding fitting portion **500** is located in the inner space **410** such that the stick portion **120** having the set diameter can be fitted between the insertion coupling portions **400**.

(57) In addition, coupling of the stick portions **120** is achieved by the protruding fitting portion **500** in the state in which the number of the stick portions **120** that can be fitted into and coupled to an arbitrary position in the inner space **410** is optionally set to one to a plurality.

(58) FIG. **6** is a view showing another embodiment of the protruding fitting portion **500a**. Referring to FIG. **6**, the protruding fitting portion **500a** may be provided at an outer circumferential surface thereof with a contact recess **510** formed in a longitudinal direction.

(59) The contact recess **510** may be formed in an arc sectional shape so as to correspond to the shape of a part of the outer circumferential surface of the stick portion **120** such that part of the outer circumferential surface of the stick portion **120** is seated in the contact recess in a contact state, and may guide the stick portions **120** so as to be located at the set positions.

(60) The depth of the contact recess **510** may be set in consideration of the positioning function, etc. of the stick portion **120**, and the number of contact recesses may be variously set.

(61) Although not shown, the protruding fitting portion **500** may be provided with a spiral protrusion formed as the result of being spirally wound a plurality of times along the outer circumferential surface thereof. Similarly to the catching portion **710**, when the cover body **350** is rotated in a state of being screwed to the specimen preservation container **200**, the protrusion **720** of the stick portion **120** may be guided along the spiral protrusion, and the spiral protrusion may be caught by the protrusion **720** of the stick portion **120** in a contact state, whereby it is possible to prevent separation of the stick portion **120**.

(62) Although not shown, the protruding fitting portion **500** may have a convex catching protrusion formed along an outer circumferential surface of an exposed end thereof, whereby it is possible to prevent separation of the inserted stick portion **120**.

(63) The separation prevention portion **700** serves to prevent the specimen collection swab **100** from being separated from the cover body **350**.

(64) Referring to FIGS. **7** and **8**, the separation prevention portion **700** is provided at the insertion coupling portion **400** so as to be caught by the stick portions **120** in a contact state when the cover body **350** is coupled to the specimen preservation container **200** to prevent the stick portions **120** from being separated from the insertion coupling portion **400**.

(65) When the cover body **350** is coupled to the specimen preservation container **200**, the separation prevention portion **700** is caught by the stick portion **120** in a contact state, whereby, when the container cover **300** for specimen storage is separated from the specimen preservation container **200**, separation of the stick portions **120** from the insertion coupling portion **400** may be prevented.

(66) The separation prevention portion **700** includes a catching portion **710** formed at an inner surface of the insertion coupling portion **400** so as to protrude therefrom and formed along the inner surface thereof.

(67) When the stick portion **120** is fitted into the insertion coupling portion **400**, the catching portion **710** is caught by a protrusion **720** of the stick portion **120**, which will be described later, whereby separation of the stick portion **120** from the insertion coupling portion **400** may be prevented.

(68) As shown, the catching portion **710** may be formed in the form of a recess, into which the protrusion **720** is inserted so as to be caught thereby, instead of protruding from the insertion coupling portion **400**, and may also be formed in various shapes, such as forming the protrusion **720** in the shape of a recess and forming the catching portion **710** in the shape of a protrusion, as long as the above purpose can be achieved.

(69) When the cover body **350** is screwed to the specimen preservation container **200**, the stick portion **120** may be guided toward the lower surface of the cover body **350** along the catching portion **710** of the separation prevention portion **700** so as to be fitted into and coupled to the insertion coupling portion.

(70) To this end, the catching portion **710** may be formed so as to be spirally wound a plurality of times around the inner circumferential surface of the insertion coupling portion **400**.

(71) This is based on consideration of the rotation of the cover body **350** and the insertion coupling portion **400** when the container cover **300** for specimen storage and the specimen preservation container **200** are screwed to each other, wherein the catching portion **710** is spirally formed in response to the rotation of the cover body **350** by screw coupling, whereby, when the insertion coupling portion **400** is rotated, the protrusion **720** is guided along the catching portion **710** toward the bottom portion **320** so as to be fitted into and coupled to the insertion coupling portion.

(72) Meanwhile, the stick portion **120** may include a protrusion **720** formed on an outer surface of the other end thereof so as to be caught by the catching portion **710** in a contact state.

(73) Here, the protrusion **720** may be formed along an outer circumferential surface of the stick portion **120** so as to annularly protrude therefrom, and a plurality of protrusions may be spaced apart from each other in a longitudinal direction of the stick portion **120**.

(74) The protrusion **720** may be formed in a hemispherically convex shape in a direction in which the stick portion **120** is inserted, as shown. However, this is merely an embodiment, and the protrusion may be formed in various shapes.

(75) Although not shown, the separation prevention portion **700** may further include a separation prevention protrusion formed as the result of being spirally wound a plurality of times along an outer circumferential surface of the protruding fitting portion **500**. Similarly to the catching portion **710**, when the container cover **300** for specimen storage is rotated, the separation prevention protrusion may be caught by the protrusion **720** in a contact state while guiding the protrusion, whereby it is possible to prevent separation of the stick portion **120**.

(76) FIGS. **9** to **11** are views showing fitting and coupling of the specimen collection swab **100** by

rotation of the container cover **300** for specimen storage.

(77) Referring first to FIG. **9**, when the container cover **300** for specimen storage is coupled to the specimen preservation container **200** so as to shield the insertion hole **230**. At this time, upper ends of the stick portions **120** of the plurality of specimen collection swabs **100** inserted into the specimen preservation container **200** are fitted between the insertion coupling portion **400** and the protruding fitting portion **500**.

(78) FIG. **10** shows that the inspector rotates the container cover **300** for specimen storage to couple the container cover **300** for specimen storage to the specimen preservation container **200** by screw coupling in the state of FIG. **9**.

(79) At this time, as the container cover **300** for specimen storage is rotated in a state of being fastened to the specimen preservation container **200** by screw coupling, the container cover for specimen storage moves downward toward the specimen preservation container **200**.

(80) As a result, the stick portion **120** is inserted more deeply in the inner space **410** toward the bottom portion **320**. Here, the protrusion **720** is guided toward the bottom portion **320** along the spiral catching portion **710** by rotation of the insertion coupling portion **400**.

(81) FIG. **11** shows the state in which the container cover **300** for specimen storage of FIG. **10** is further rotated. Referring to the figure, the container cover **300** for specimen storage is further rotated such that the open end of the specimen preservation container **200** is located adjacent to the bottom portion **320** of the cover body **350**, whereby the container cover for specimen storage is airtightly coupled to the specimen preservation container **200** in tight contact therewith.

(82) The stick portion **120** is further inserted toward the bottom portion **320** in the insertion coupling portion **400**, and the protrusions **720** of the stick portion **120** are caught by the catching portion **710**, whereby separation of the stick portion **120** from the insertion coupling portion **400** is prevented.

(83) The catching portion **710** is spirally formed on the inner circumferential surface of the insertion coupling portion **400**, as described above. When the container cover **300** for specimen storage is rotated by screw coupling, therefore, the catching portion may guide the protrusions **720**, whereby easier catching contact may be achieved.

(84) When the inspector separates the container cover **300** for specimen storage from the specimen preservation container **200** for inspection in this state, the specimen collection swab **100** is removed in a state of being coupled to the insertion coupling portion **400**, and therefore it is not necessary for the inspector to separately remove the specimen collection swab **100** from the specimen preservation container **200**.

(85) In a specimen collection and storage kit according to an embodiment of the present invention, it is very important to secure airtightness in a process of storing and transporting a specimen collected from a subject in order to obtain effectiveness of a test by preventing leakage of a specimen and a medium as well as contamination from the outside.

(86) In the present invention, therefore, an airtight portion **800** may be formed at each of the container cover **300** for specimen storage and the specimen preservation container **200**, which are coupled to each other, in order to further improve airtightness.

(87) When the specimen preservation container **200** is fitted into and coupled to the interstitial space **330** between the inner circumferential surface of the cover body **350** and the outer surface of the insertion coupling portion **400**, the airtight portion **800** may be formed at each of the inner circumferential surface of the side portion **310** and the outer surface of the insertion coupling portion **400**, whereby, when the specimen preservation container **200** is fitted into and coupled to the interstitial space **330**, the specimen preservation container **200** and the cover body **350** are brought into tight contact with each other, and therefore airtightness may be maintained.

(88) Referring to FIGS. **12** and **13**, the airtight portion **800** may include a first pressing protrusion **810** and a second pressing protrusion **820**.

(89) First, the first pressing protrusion **810** may be formed at the inner circumferential surface of

the side portion **310** of the cover body **350** so as to protrude therefrom in order to press an outer circumferential surface of one end of the specimen preservation container **200** when the specimen preservation container **200** and the container cover **300** for specimen storage are fastened to each other. Here, the first pressing protrusion **810** may be located adjacent to the bottom portion **320** to press the end of the specimen preservation container **200**.

(90) The second pressing protrusion **820** may be formed at the outer circumferential surface of the insertion coupling portion **400** so as to protrude therefrom in order to press an inner circumferential surface of one end of the specimen preservation container **200** when the specimen preservation container **200** and the container cover **300** for specimen storage are fastened to each other. In this case, the second pressing protrusion **820** may be located adjacent to the bottom portion **320** of the container cover **300** for specimen storage to press the end of the specimen preservation container **200**.

(91) The first pressing protrusion **810** and the second pressing protrusion **820** may be located so as to be staggered from each other, as shown; however, the positions thereof may be changed. For example, the first pressing protrusion and the second pressing protrusion may be located at the same position on a horizontal line.

(92) Each of the first pressing protrusion **810** and the second pressing protrusion **820** may be formed such that one side thereof is inclined in an insertion direction, as shown, to guide the insertion of the specimen preservation container **200**.

(93) According to the foregoing, the airtight portion **800** may press the inside and the outside of one end of the specimen preservation container **200** through the first pressing protrusion **810** and the second pressing protrusion **820**, respectively, and block any space that may occur between the specimen preservation container **200** and the container cover **300** for specimen storage, whereby it is possible to improve sealing force and airtightness.

(94) Meanwhile, edges of the specimen preservation container **200** and the container cover **300** for specimen storage have a small area of contact as the edges are in line contact rather than in surface contact, and it is necessary to secure airtightness of the edges as the edges may be separated from each other.

(95) Accordingly, the airtight portion **800** may include a first inclined contact surface **831** and a second inclined contact surface **832** to secure airtightness at the edge of the specimen preservation container **200** in addition to the first pressing protrusion **810** and the second pressing protrusion **820** provided to maintain airtightness at the inner surface and the outer surface of the specimen preservation container **200**.

(96) First, the first inclined contact surface **831** may be formed such that an end edge of the specimen preservation container **200** is incised to have an inclined surface.

(97) The second inclined contact surface **832** may be formed at a lower end of the insertion coupling portion **400** that faces the first inclined contact surface **831**, and may be formed so as to have an inclination corresponding to the inclination of the first inclined contact surface **831**.

(98) When the specimen preservation container **200** is inserted into and coupled to the interstitial space **330**, therefore, the first inclined contact surface **831** and the second inclined contact surface **832** come into tight surface contact with each other, whereby airtightness at the edge thereof is secured.

(99) Although the first inclined contact surface **831** and the second inclined contact surface **832** are shown as being formed at the inner circumferential end edge of the specimen preservation container **200**, the first inclined contact surface and the second inclined contact surface may be formed at the outer circumferential end edge of the specimen preservation container **200** or at each of the inner and outer circumferential end edges of the specimen preservation container.

(100) Furthermore, the airtight portion **800** may include a protrusion **220** formed on an outer circumferential surface of one end of the specimen preservation container **200** in tight contact with the side portion **310** of the cover body **350**. The protrusion **220** may be formed at an outer

circumferential surface of an upper end of the specimen preservation container **200** so as to annularly protrude therefrom to press the side portion **310** for tight contact.

(101) A plurality of protrusions **220** may be arranged from the upper end of the specimen preservation container **200** in a longitudinal direction so as to be spaced apart from each other, and the number of the protrusions and the distance between the protrusions may be variously adjusted.

(102) FIGS. **14** and **15** are views showing the state in which the specimen preservation container **200** and the container cover **300** for specimen storage are fastened to each other and the airtight portion **800**.

(103) Referring to the figures, the airtight portion **800** presses the inner circumference surface and the outer circumferential surface of the specimen preservation container **200** through the first pressing protrusion **810** and the second pressing protrusion **820**, respectively, such that the ends of the specimen preservation container **200** and the container cover **300** for specimen storage come into tight contact with each other.

(104) In addition, the airtight portion **800** brings the edges of the specimen preservation container **200** and the container cover **300** for specimen storage into tight contact with each other through the first inclined contact surface **831** and the second inclined contact surface **832**, whereby overall airtightness may be improved.

(105) The container cover **300** for specimen storage may be made of an elastically deformable material capable of resiliently pressing the specimen storage container **200** fitted into the interstitial space **330**, and various materials, including resin-based materials, may be used as the material of the container cover for specimen storage.

(106) FIG. **16** is a view showing another embodiment of the container cover for specimen storage **300a**. Referring to the figure, the container cover **300a** for specimen storage may be configured such that the bottom portion **320** is formed concave toward said receiving space **210** and the protruding fitting portion **500** is formed up to the end of the insertion coupling portion **400**. In the specimen collection and storage kit according to the present invention, as described above, the shape of the container cover **300a** for specimen storage and the shape, diameter, and length of the protruding fitting portion **500** may be variously changed as long as the purpose of each of the above components can be achieved.

(107) FIG. **17** is a view showing a use example of the container cover **300** for specimen storage according to the embodiment of the present invention. Referring to the figure, the inspector first collects a specimen from a subject through the specimen collection swab **100** and inserts the specimen collection swab into the receiving space **210** of the specimen preservation container **200**.

(108) Subsequently, the inspector couples the container cover **300** for specimen storage to the open one side of the specimen preservation container **200** to shield the specimen preservation container **200**. At this time, the stick portion **120** of the specimen collection swab **100** is fitted into the insertion coupling portion **400**.

(109) At this time, the inspector rotates the container cover **300** for specimen storage so as to be screwed to the specimen storage container **200**. In this process, the stick portion **120** is further inserted inwardly of the insertion coupling portion **400**. At this time, the protrusions **720** are guided inwardly of the insertion coupling portion **400** along the catching portion **710** and are caught by the catching portion **710** in a contact state, as described above.

(110) Subsequently, when the inspector separates the container cover **300** for specimen storage, the specimen collection swab **100** comes out in a state of being fitted in the container cover **300** for specimen storage by the separation prevention portion **700**, as shown.

(111) The invention has been described with reference to the embodiment shown in the drawings, but this is illustrative only, and a person having ordinary skill in the art to which the present invention pertains will understand that various modifications and other equivalent embodiments are possible therefrom. The true scope of technical protection of the present invention will therefore be determined by the technical ideas of the appended claims.

(112) The present invention is applicable to a specimen collection instrument.

Claims

1. A container cover for specimen storage, the container cover comprising: a cover body comprising a tubular side portion and a bottom portion coupled to one end of the tubular side portion, the cover body being detachably screwed to open one side of a specimen preservation container in which one or more specimen collection swabs are provided by insertion, each of the one or more specimen collection swabs including a stick portion; an insertion coupling portion formed so as to protrude from the bottom portion toward the specimen preservation container, the insertion coupling portion having an inner space defined therein such that the stick portion can be inserted and fitted into and coupled to the inner space; and a protruding fitting portion located in the inner space, the protruding fitting portion being formed so as to protrude from the bottom portion, the protruding fitting portion being configured such that the stick portion is fitted and coupled between an outer surface of the protruding fitting portion and an inner surface of the insertion coupling portion, wherein when the specimen preservation container is coupled to the container cover, the stick portion is fitted into and coupled to a position in the inner space, the container cover further comprising a separation prevention portion provided at the insertion coupling portion, the separation prevention portion being caught by the stick portion in a contact state when the cover body is coupled to the specimen preservation container to prevent the stick portion from being separated from the insertion coupling portion, wherein the separation prevention portion comprises a catching portion formed at the inner surface of the insertion coupling portion so as to protrude therefrom and formed along the inner surface, when the cover body is screwed to the specimen preservation container, the stick portion is guided toward a lower surface of the cover body along the catching portion so as to be fitted into and coupled to the insertion coupling portion, the catching portion is formed so as to be spirally wound a plurality of times around the inner surface of the insertion coupling portion in response to rotation of the container cover by screw coupling, and when the cover body is screwed to the specimen preservation container, the stick portion is guided toward the lower surface of the cover body along the spiral catching portion so as to be fitted into and coupled to the insertion coupling portion.
2. The container cover according to claim 1, wherein the insertion coupling portion is formed in a shape of a cylindrical tube, and the stick portion is inserted into a position in the cylindrical inner space.
3. The container cover according to claim 1, wherein the insertion coupling portion is configured such that an exposed end thereof is formed so as to be inclined downward toward an inside thereof to guide the stick portion.
4. The container cover according to claim 1, wherein the protruding fitting portion is formed in a cylindrical shape.
5. The container cover according to claim 1, wherein the specimen preservation container is fitted into and coupled to an interstitial space between an inner circumferential surface of the cover body and an outer surface of the insertion coupling portion, and the container cover further comprises an airtight portion formed at each of an inner circumferential surface of the tubular side portion and the outer surface of the insertion coupling portion, the airtight portion being configured to allow the specimen preservation container and the cover body to come into tight contact with each other such that airtightness is maintained when the specimen preservation container is fitted into and coupled to the interstitial space.
6. The container cover according to claim 5, wherein the airtight portion comprises: a first pressing protrusion formed at the inner circumferential surface of the tubular side portion of the cover body so as to protrude therefrom, the first pressing protrusion being configured to press an outer

circumferential surface of the specimen preservation container; and a second pressing protrusion formed at an outer circumferential surface of the insertion coupling portion so as to protrude therefrom, the second pressing protrusion being configured to press an inner circumferential surface of the specimen preservation container.
